

Bags, Vectors & Transformers

Python & Colab · Day 1, Part 2

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What is Python?

Python is a **general-purpose programming language** that has become the **dominant tool** in data science and research. It is **free, open source**, and **readable**. For **computational text analysis** it is the standard choice — and although the examples here are in English, the same libraries (spaCy, HuggingFace) support **many other languages**, including Dutch.

Why researchers use it

- **Free** and **open source**
- **Readable**, close to plain English
- **Huge community** and documentation
- Runs on **any operating system**

Key libraries for text analysis

- **NLTK**, **spaCy** — preprocessing (spaCy: many languages)
- **scikit-learn** — BoW, TF-IDF
- **Gensim** — topic modeling, embeddings
- **HuggingFace** — transformers, BERT

Python vs. R

Many of you probably already know R. Both are valid — here is how they compare for this kind of work.

Python tends to lead on...

- **Deep learning** and **transformers**
- Working with **embeddings** and **LLMs**
- **End-to-end pipelines** (collect, clean, analyze)
- Most **new NLP methods** appear here first

R tends to lead on...

- **Statistical modeling** and survey analysis
- **Tidy text** workflows (tidytext, quanteda)
- **Visualization** (ggplot2)
- Widely used in **social science** departments

If you know R, Python will feel familiar quickly. This course uses Python, but the concepts transfer.

What is Google Colab?

Google Colab is a **free, browser-based environment** for writing and running Python code. **No installation needed** — you open a link and start coding. Everything runs on **Google's servers**, not your own machine. It uses the **notebook format**: code cells you run one at a time, mixed with text cells for notes. Great for teaching, and for keeping analysis **readable and reproducible**.

Quick facts

- Runs entirely in your **browser**
- **Free tier** available
- Access to **GPU** for heavier models
- **Easy to share**, like a Google Doc
- Built on **Jupyter Notebooks**

How we will use it

- All course notebooks **shared as links**
- Open, run, experiment
- **Save a copy** to your own Drive
- You will need a **Google account**

Colab: Pros and Cons

Pros

- **Zero setup**, works out of the box
- **Same environment** for everyone
- Free **GPU access** for heavier models
- Easy to **share and collaborate**
- All course materials in **one link**
- **Saves automatically** to Google Drive

Cons

- Needs a **stable internet** connection
- Sessions **time out** after inactivity
- Free tier has **usage limits**
- Not great for **very large datasets**
- **Data privacy** considerations (Google)
- **Less control** than a local environment

For this course, the pros far outweigh the cons. Once you are comfortable, setting up a local environment is a natural next step.